GEODYNAMIC INTERPRETATIONS FROM PALEOMAGNETIC DATA OF LATE PALEOZOIC ROCKS IN THE SOUTHERN ANDES

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Abstract. Paleomagnetic data for three Late Paleozoic formations from the Argentine Andes are reported. These formations are: the Majaditas Formation, exposed at 31.8°S 69.4°W (Precordillera) assigned to the Middle Carboniferous; the Portezuelo del Cenizo Formation, exposed at 32.1°S 69.5°W (Cordillera Frontal) with a radiometric age of 275±10 Ma (Early Permian) and the Choiyoi Formation, exposed at 39.1°S 70.5°W (Cordillera Principal), assigned to the Permo-Triassic. The directions of the stable remanent magnetization (srm) isolated for the rocks of these formations were submitted to tilt tests. In every case these tests indicated that the srm isolated was acquired before the strata were tilted.

The paleomagnetic poles (PP) computed for the three formations (Majaditas Fm.: 200°E 42°S, A95=8°; Portezuelo del Cenizo Fm.: 212°E 30°S, A95=7°; Choiyoi Fm.: 232°E 21°S, A95=8°) are not coincident with those for coeval rocks exposed in stable areas of South America. This is interpreted in terms of two different hypotheses. The first suggests that the sampling areas of these formations are situated in crustal blocks that suffered different large clockwise rotations around vertical axes during the Mesozoic or Cenozoic. The other suggests that these rocks together with the Late Carboniferous Yalguaraz Formation, exposed in Cordillera Frontal, were part of a displaced terrane - the limits of which are not yet defined - that was accreted to the western edge of South America prior to the Early Triassic.

Introduction

Several authors have recently suggested the possible existence of allochthonous terranes in southwestern South America (Nur and Ben Avraham, 1977,1978; Martinez, 1980; Dalmayrac et al, 1980; Ramos, 1984; Ramos et al, 1984, 1986). Particularly, Vilas and Valencio (1982; see also Valencio and Vilas, 1985) carried out a paleomagnetic study of some late Paleozoic rocks from the Cordillera Frontal, Argentina (Yalguaraz Formation, Upper Carboniferous? and Portezuelo del Cenizo Formation, Early Permian; sampling area 2, Figure 1). The paleomagnetic pole positions obtained for these formations differ considerably from those for age-equivalent rocks from stable zones of South America. These authors interpreted these results in terms of two different hypotheses. One suggested a large "in situ" clockwise rotation of site 2 around a vertical axis after Permian times. The other postulated that these rocks are part of a displaced terrane which accreted to the western continental margin of South America in the Permo-Triassic.

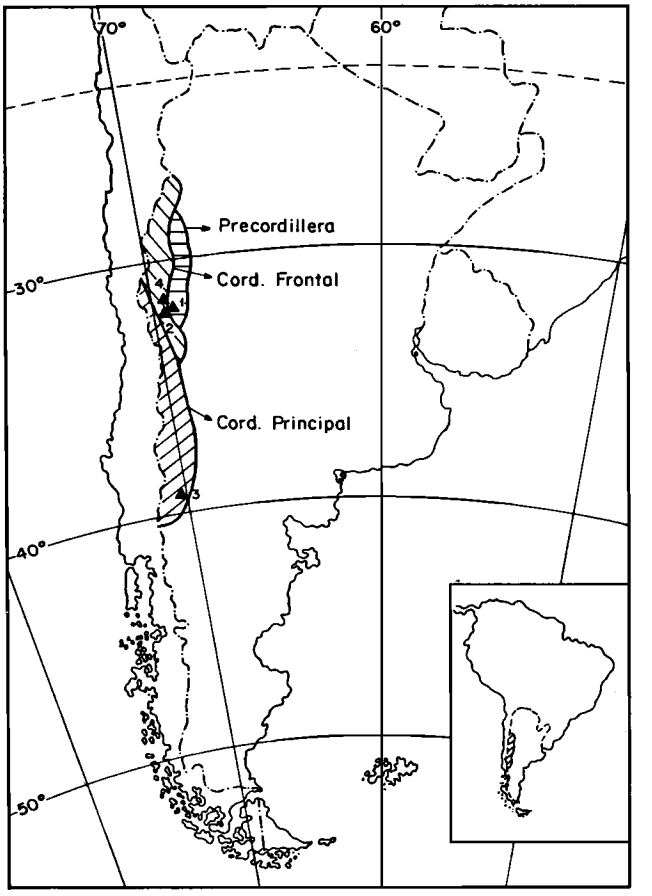


Fig.1. Location of the sampling areas. 1) Majaditas Formation; 2) Yalguaraz and Portezuelo del Cenizo Formations; 3) Choiyoi Formation; 4) Ignimbrites of the Choiyoi Group. The Precordillera, Cordillera Frontal and Cordillera Principal morphotectonic provinces are also shown.

In order to test these interpretations, paleomagnetic studies were carried out on Middle Carboniferous sedimentary rocks of the Majaditas Formation, exposed in the Western Precordillera, Argentina (sampling area 1, Figure 1), and on the Permo-Triassic volcanic and pyroclastic rocks of the Choiyoi

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Formation, exposed at Cerro Chachil and Sierra de Catan Lil, in the Cordillera Principal, Argentina (sampling area 3, Figure 1). A complementary study on rocks from the Portezuelo del Cenizo Formation was also carried out to confirm the paleomagnetic results mentioned above (Vilas and Valencio, 1982).

Geologic Setting and Paleomagnetic Studies

Majaditas Formation

Geology. Figure 2 is a schematic geologic map of the Uspallata - Calingasta Valley region. The Majaditas Formation (Amos and Rolleri, 1965) is exposed at sampling area 1 (31.8°S, 69.4°W), in the Western Precordillera, 15 km southeastward from Barreal, in San Juan Province, Argentina. This formation is composed of a thick sequence (650 meters) of clastic sedimentary rocks ranging from claystones to conglomerates that uncomformably overlays a highly deformed preCarboniferous sequence of marine claystones and sandstones. The upper section, which was sampled for the paleomagnetic study, consists mainly of red sandstones and fine conglomerates which grade upward from turbidite to littoral facies. No fossils have been found in the Majaditas Formation. Nevertheless, a Westphalian age has been assigned to it on the basis of stratigraphic and lithologic correlation with other similar Carboniferous formations of the Western Precordillera (Lopez Gamundi and Alonso, 1982).

The Majaditas Fm. as well as the preCarboniferous rocks are intruded, in this area, by several dykes, sills and other shallow intrusive bodies. In the sampling area, it is exposed in a north - south trending homocline with a westward dip of about 60 to 70 degrees. The upper fifty meters of the sequence are overturned, probably due to deformation associated with a north trending thrust fault named "Puesto de Tapia Fault" (Quartino, Zardini and Amos, 1971).

Paleomagnetic study. Paleomagnetic data for the upper 205 meters of the Majaditas Fm. are reported. Forty-one oriented hand samples from different stratigraphic levels were collected at the sampling area 1 (Fig.2). The natural remanent magnetization (nrm) of two specimens from each sample was measured in order to establish the within sample consistency of direction of the remanent magnetization. Fifteen pilot samples (two specimens for each sample) were submitted to progressive thermal and chemical demagnetization treatments (Irving, 1964) in order to isolate the stable remanent magnetization (srm). Thermal cleaning stages of 100, 200, 300, 375, 450, 500, 550, 600, 650 and 660°C were applied. Hydrochloric acid (7N) at 70°C was used for the chemical demagnetization. In this case the stages lasted generally 1, 2, 4, 6, 24, 48, 96, and sometimes 144 and 288 hours respectively. The srm directions isolated by both cleanings techniques were averaged for each pilot sample. As both treatments resulted in similar demagnetization behaviour (Fig.3), the specimens from the remaining twenty six samples were only submitted to thermal cleaning with stages at 200, 300, 400, 500 and in some cases 600°C.

Srm were isolated for 35 of the 41 samples. About 75% of these carried at least two different superimposed magnetic components. One of these components was interpreted as the srm, while the others were interpreted as secondary magnetizations and were easily removed by the demagnetization treatments (e.g. sample M-109 in Fig.3a). These samples showed in general Curie points near 575°C, which suggests that magnetite is the dominant ferromagnetic phase. The remaining 25% carried only a single magnetic component with negative or positive inclination, the srm (e.g. sample M-133 in Fig.3b). These samples showed high stability until 600°C. Over this temperature all of them increased their susceptibility from two to ten times becoming their magnetic remanences unstable in direction and intensity, which is probably indicating chemical changes induced by the heating. Curie temperatures beyond 600°C suggest that hematite can be the dominant ferromagnetic phase in these samples. Although the Curie points could not be reached in these cases, the direction of the srm was confirmed by the chemical leaching (specimen M-133b, Fig.3b).

Samples from the Majaditas Fm. were collected from sequences in two tilted blocks (sites A and B, Table 1). This allowed us to apply a tilt test (McElhinny, 1964; McFadden and Jones, 1981) to the srm directions isolated. The "in situ" srm directions from sites A (circles) and B (triangles), and the means for each one are shown in Figure 4a. Note that both mean directions have a small overlap before the structural corrections are applied. Instead, after applying them, both overlap perfectly (Fig.4b). This means that the srm isolated for the Majaditas Formation is pretilting. Nevertheless, Fig.4b shows that the srm directions do not pass a reversal test. Those directions with shallower inclinations - all the directions in the NE quadrant and those from samples M99 and 101 in the third - correspond to the univectorial samples (hematite?) while the others correspond to the multivectorial ones (magnetite?). This suggests that the failure of the reversal test may be due to the fact that the acquisition of the magnetization is probably related to a different process in each case - sedimentation for the magnetite and diagenesis for the hematite? - although we are not certain of it.

A virtual geomagnetic pole (VGP) was then computed for each sample (two specimens) (Table 1). They showed both reversed and normal polarities. The mean is the paleomagnetic pole (PP) for the upper section of the Majaditas Formation (C8): 200°E, 42°S, N=31, k=12.6, A95=8° and e= 0.646. e is an statistical parameter that indicates the ellipticity of the VGPs distribution (Engebretson and Beck, 1978). All the VGPs located beyond 40° from the mean or its antipode were considered recordings of transitional fields or excursions and excluded from the average (marked with * in Table 1), following the criteria established by McElhinny et al (1974) on the basis of the characteristics of the dipole moment (Dagley et al, 1967; Lawley, 1970; Wilson et al, 1972).

If the VGPs correspondent to the univectorial (hematite?) samples are not taken into account, the PP of the Majaditas Fm. is: 210°E, 47°S, N=20, k=14.3, A95=9° and e= 0.862. It clearly follows that the positions of both PP are close and that their precision parameters k are similar, although the distribution of VGPs in the first case is rather less elliptical.

Portezuelo del Cenizo Formation

Geology. The location of the sampling area 2 (32.1°S, 69.5°W) is shown in Figure 2. This figure also includes a more detailed geologic map of this sampling area which shows the location of the sampling sites A (Yalguaraz Formation) and B (Portezuelo del Cenizo Formation) from the paleomagnetic study carried out by Vilas and Valencio (1982); and the site C which corresponds to the complementary study of the Portezuelo del Cenizo Fm. reported here.

The oldest rocks exposed in this area are the highly deformed and mildly metamorphosed marine sedimentary rocks of the Cienaga del Medio Group (Amos and Marchese, 1964). These rocks have been assigned to the Devonian and are overlain by the clastic marine sedimentary rocks of the Yalguaraz Fm. (Amos and Rolleri, 1965) which has been recently assigned to the Upper Carboniferous on the basis of paleontological evidence (Taboada, 1985). A marked angular uncomformity has been recognized between them. The samples studied by Vilas and Valencio (1982) were collected from the upper thirty meters of the Yalguaraz Fm. at site A. The contact between this formation and the overlying Portezuelo del Cenizo Fm. (Coira and Koukharsky, 1976) is slightly uncomformable. The latter consists in a calcalkaline volcanic and pyroclastic sequence with a maximum thickness of 300 meters. According to Cortes (1985), this formation is the base of the Choiyoi Group (Stipanicic, 1965) in this region. This Group includes almost all the Permo-Triassic volcanic and associated pyroclastic rocks from the Argentine Andes. The Portezuelo del Cenizo Fm. includes a lower member, composed of andesitic breccias and conglomerates, a middle member of interbedded dacitic tuffs and breccias, and an upper member, composed of porphyritic andesites and andesitic breccias. A K/Ar radiometric date of 275±10 m.a. (Vilas and Valencio, 1982) indicates an Early Permian age for this formation, which is overlain by the thick rhyolitic and ignimbritic Permo-Triassic sequence of the Horcajo Formation (Mirre,

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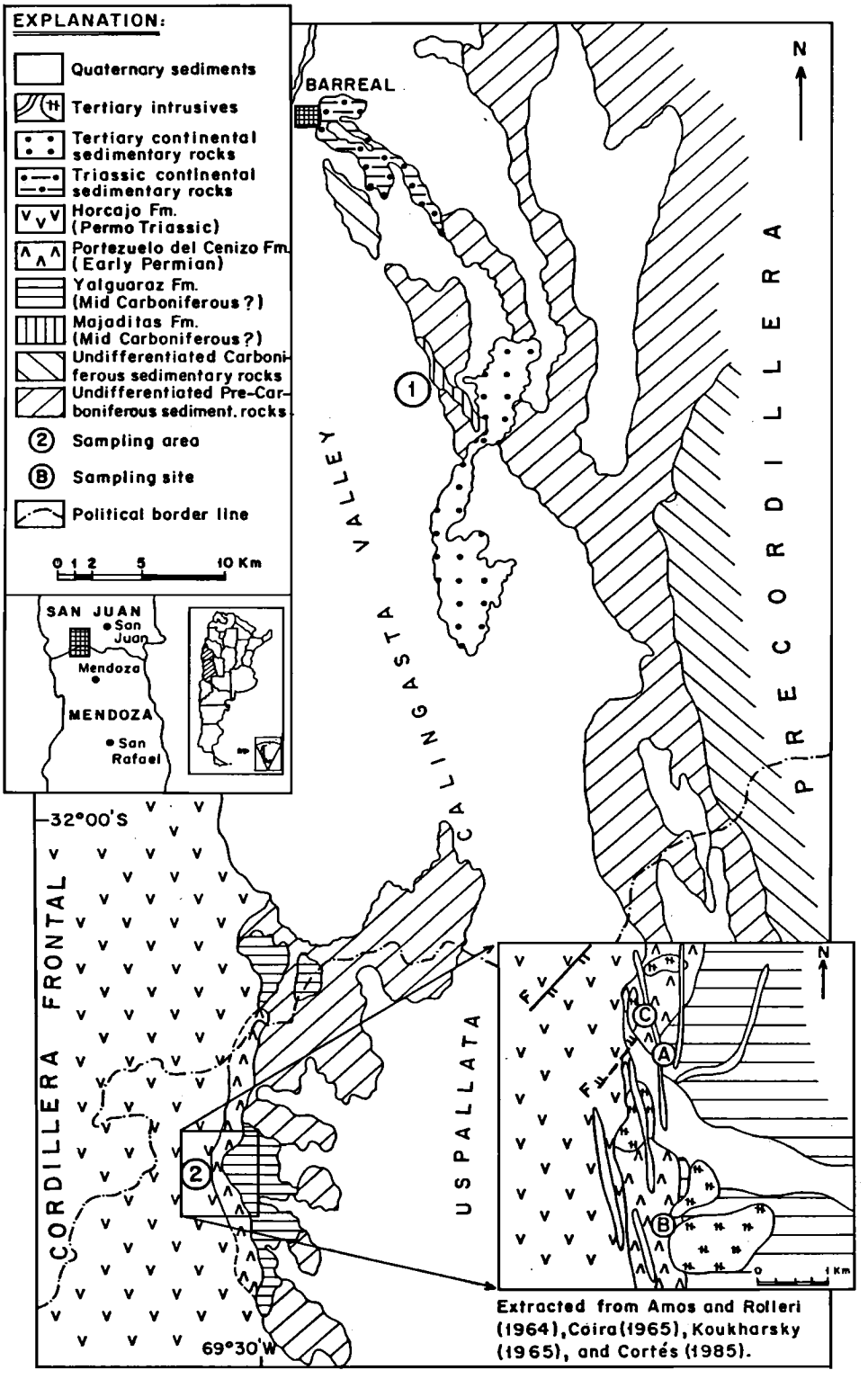


Fig.2. Geologic map of the Uspallata - Calingasta Valley region and the location of the sampling areas 1 and 2. A detailed geologic map of the sampling area 2 with the location of the sampling sites A, B and C is also included.

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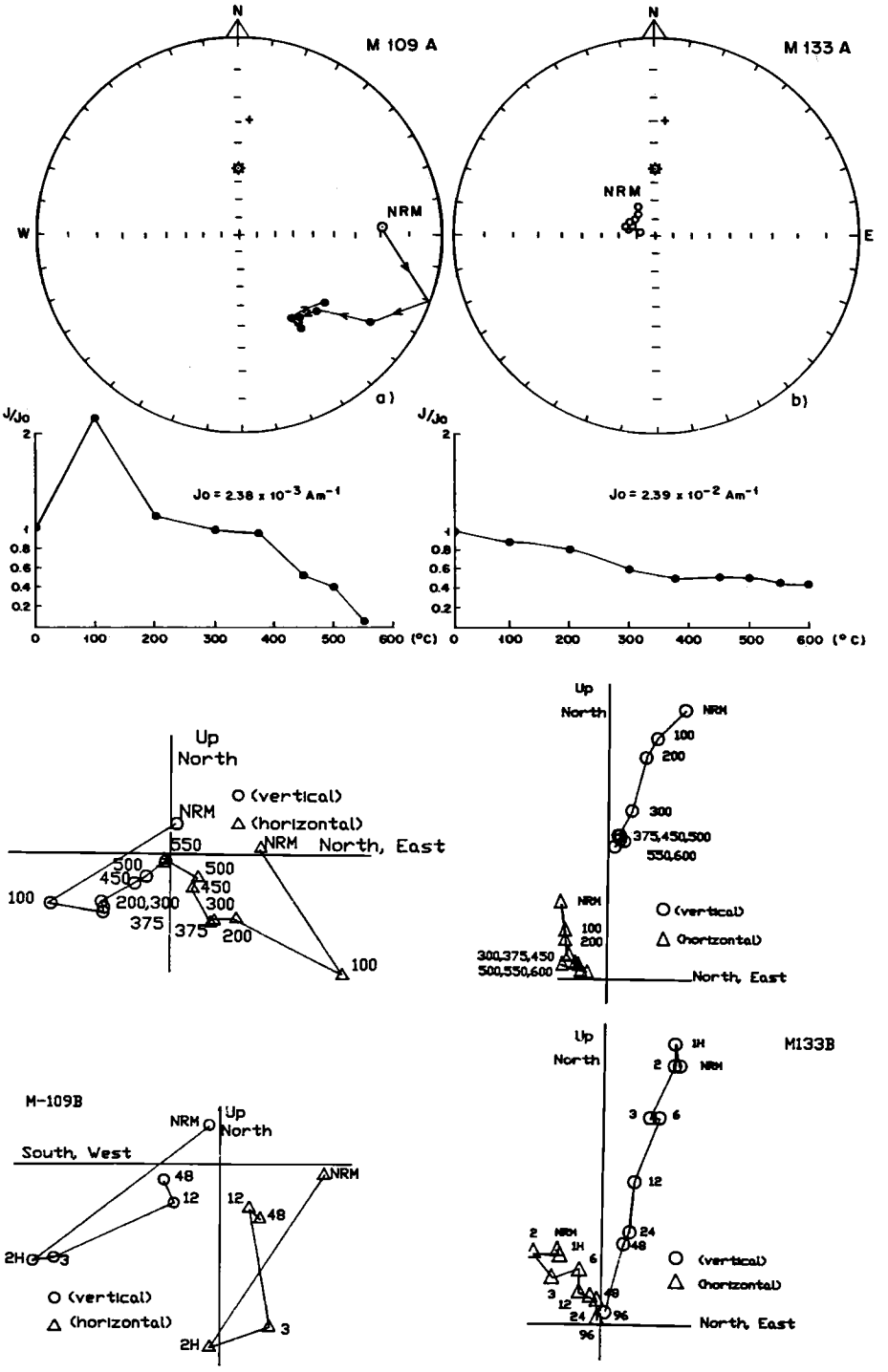


Fig.3. Typical magnetic behaviour of specimens from the Majaditas Formation submitted to thermal cleaning. The direction of the remanent magnetization (not corrected for tilt) and the relative intensity after each cleaning stage are plotted in a paired stereonet - demagnetization curve and in an orthogonal vector diagramm. The magnetic behaviour of other specimens of the same samples submitted to chemical demagnetization is shown only in orthogonal vector diagramms. Open circles: negative inclination; solid circles: positive inclination; + : present geomagnetic field direction; spiked circle: axial dipole field direction. NRM: natural remanent magnetization.

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1966), also part of the Choiyoi Group. The contact between them seems to be conformable. All the formations mentioned above are commonly intruded by Tertiary andesitic and rhyolitic porphyries and dykes.

In this area, the late Paleozoic rocks are exposed in a homoclinal which is often disrupted by faults that have caused local tilting of several crustal blocks. This allowed us to carry out a complementary paleomagnetic sampling on rocks from the Portezuelo del Cenizo Formation (site C) where these rocks have a clearly different attitude from that present at site B (Table 2).

Paleomagnetic study. Vilas and Valencio (1982) sampled the middle member of the Portezuelo del Cenizo Formation at site B. A complementary study, reported in this paper, was performed on fifteen hand-samples collected from eight andesitic flows of the upper member at site C. Two specimens from each sample were submitted to alternating field demagnetization (Irving, 1964). In general, at least seven cleaning stages were applied in order to isolate the direction of the srm. Magnetic fields as high as 90 mT were applied to some specimens to test their magnetic stability. Ten of the 15 samples showed a single magnetic component with a positive inclination and high coercivities (e.g. specimen AT-6A in Fig. 5a). The other five, sampled from the lowest two flows of the sequence, also carried a secondary magnetic component with low coercivities and coincident with the axial dipole direction (e.g. specimen AT-14A, Fig.5b). This secondary component could be related to heating caused by the intrusion of the rhyolitic and andesitic dykes and porphyries during the Tertiary.

The directions of the srm isolated for both specimens from each sample were averaged, as well as the directions obtained from multiple samples of the same flow. A unique srm direction for each flow was thus obtained (Table 2). These directions as well as their mean are plotted in Fig. 6a (circles). The srm directions and their mean obtained by Vilas and Valencio for the intermediate member are also shown there (triangles). Fig.6a clearly shows that the "in situ" mean directions for each population are not coincident and that the respective 95% confidence circles do not intersect. Instead, when the respective structural corrections are applied (Fig.6b) the mean directions become coincident. This indicates that the srm isolated for the Portezuelo del Cenizo Fm. is pretilting. A VGP was then calculated for each flow of the upper member. These VGPs together with those for the middle member are shown in Table 2. All these VGPs were then averaged, excluding all of them located beyond 40° from the mean, in order to compute the new paleomagnetic pole for the Portezuelo del Cenizo Formation. This paleomagnetic pole supersedes that published previously by Vilas and Valencio (1982). Although the statistical parameters have been improved, the position is very close to that calculated previously. The new parameters are: (P13), 212°E, 30°S, N=17, k=27.1, A95=7°.

Choiyoi Formation

Geology. Figure 7 is a geologic map of the sampling area 3 (after Neher, 1986). This region is situated in the extreme south of the Cordillera Principal (39.1°S, 70.5°W). The oldest rocks exposed here are the Lower Paleozoic metamorphic rocks of the Piedra Santa Fm. (Di Gregorio and Uliana, 1980). These rocks are intruded by the large granitic to granodioritic batholithes of the Huechulaufquen Fm. (Turner, 1965) which are considered to be Late Carboniferous - Early Permian on the basis of stratigraphic relations and a radiometric date of 281±4 m.a (Sillitoe, 1977).

The Choiyoi Fm. (Rolleri and Criado Roque, 1969) overlies these rocks,

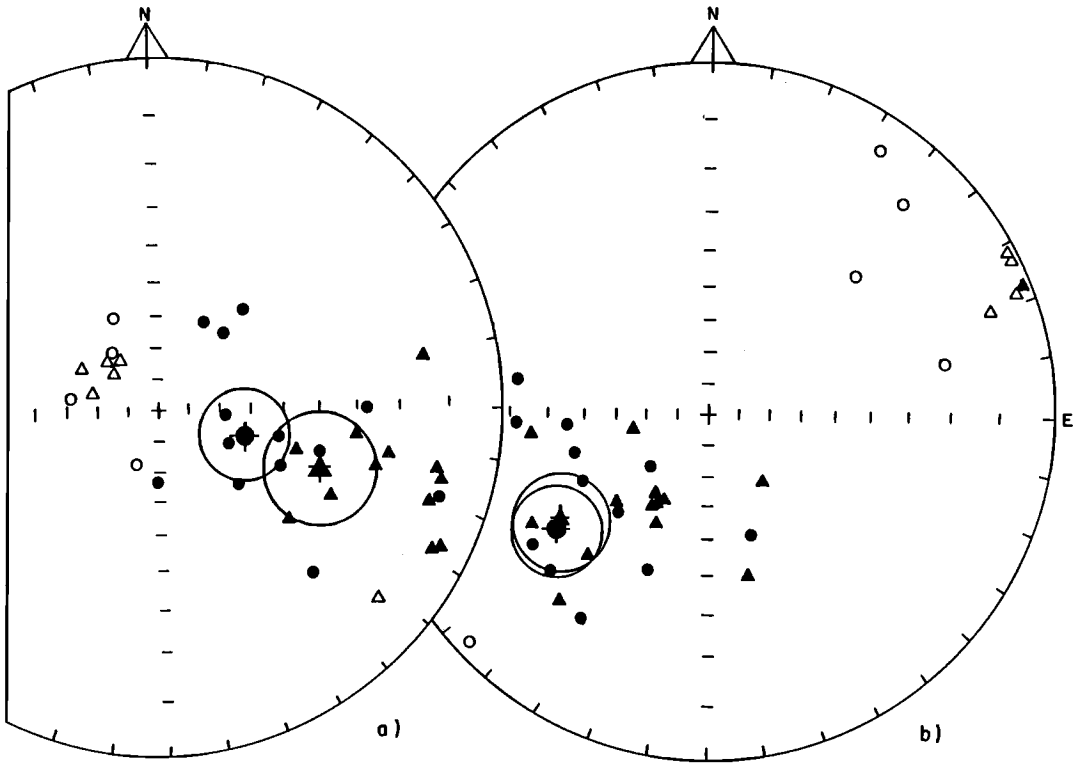


Fig.4. a) Directions of the stable remanent magnetization (srm) without structural correction for samples from the Majaditas Formation. Circles: samples from site A; triangles: samples from site B (overturned section); crossed circle and triangle: mean directions. The 95% confidence circles are also shown. Open symbols: negative inclinations; solid symbols: positive inclinations. b) Directions of the srm after the application of the structural corrections.

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TABLE 1. Paleomagnetic data for the Majaditas Formation

	sample No	Jsrm/ Jnrm (°C)	Selec. stage	Inc. (o)	SRM Dec. (o)	A95 (o)	Attitude strike (o)	dip (o)	Pre-folding SRM Inc. (o)	Dec (o)	.A95 (o)	Lat.S (o)	VGP Long.E (o)	A95 (o)	
Site A	99(2)	0.55	500	54.2	134.0		157	81W	20.6	211.8		53.8	171.9		
	100(2)	0.62	500	64.0	116.3				24.9	225.5		44.4	187.0		
	101(2)	0.58	500	66.5	183.8				-1.9	226.0		35.4	172.8		
	102(2)	0.52	300	9.0	109.1				48.4	160.4		72.9	23.3		*
	103(2)	0.20	500	38.4	107.4		168	84W	48.0	222.7		53.2	207.7		
	105(2)	0.37	400	57.9	42.5				31.2	279.1		1.4	219.5		*
	106(2)	0.80	400	27.0	91.8				65.1	227.6		50.8	237.8		
	107(2)	1.04	500	50.7	104.5		178	81W	46.1	252.9		27.6	217.0		
	108(2)	0.95	500	46.9	116.3				45.0	240.7		37.4	211.4		
	109(2)	0.43	500	23.4	138.1				39.9	201.1		68.9	180.3		
	110(2)	0.45	500	58.5	27.0		155	83W	30.8	266.9		11.1	213.0		
	111(2)	0.39	500	48.1	43.8				45.0	264.9		17.6	221.3		
	114(2)	0.70	500	-61.4	278.9				-30.0	47.0		-44.7	11.7		
	115(2)	0.48	550	-71.1	203.4				-20.7	78.1		-15.6	23.6		
	116(2)	0.70	400	-58.1	335.2				-5.8	33.0		-47.6	344.3		
	117(2)	0.88	400	-67.1	321.6				-11.6	42.2		-42.8	356.0		
	118(2)	1.38	400	67.7	97.1				25.5	232.0		39.1	191.0		
Mean Site A		N=15		60.2	108.0	13			31.9	232.1	12	41.2	198.2	11	
Site B	121(2)	0.50	300	6.3	118.5		180	102W	56.1	211.3		63.7	220.1		
	122(2)	0.30	375	5.3	117.1				57.8	211.2		63.7	223.9		
	123(2)	0.32	375	29.7	98.1		181	115W	34.8	263.4		15.3	213.8		
	124(2)	0.52	400	10.5	110.9		170	102W	51.9	205.0		68.8	209.3		
	125(2)	0.81	375	31.0	117.7				33.7	220.9		50.9	190.6		
	126(2)	1.50	400	36.9	131.3				21.3	217.9		49.3	178.5		
	128(2)	0.82	400	8.7	105.2				57.4	208.3		65.9	223.6		
	129(2)	1.40	375	13.0	80.8				64.9	258.0		30.3	239.7		
	130(2)	0.70	400	-10.7	132.1				38.9	165.8		74.0	54.7		*
	131(2)	0.45	300	44.5	107.0				28.3	238.3		34.6	197.3		
	133(2)	0.51	375	-72.7	310.6				0.8	66.7		-19.3	7.4		
	134(2)	0.90	400	-71.6	300.0				-2.1	68.3		-18.9	9.6		
	135(2)	0.77	375	-68.5	318.4				0.3	61.8		-23.5	4.6		
	136(2)	0.59	400	-67.9	285.8				-7.9	70.5		-18.6	13.4		
	137(2)	0.63	375	-68.5	314.1				-1.0	62.7		-23.2	5.8		
	138(2)	0.45	500	21.0	102.3				50.3	226.2		50.8	212.2		
	140(2)	0.20	375	10.2	104.7				53.7	211.1		63.8	221.5		
	141(2)	0.26	400	-24.7	106.6				61.7	139.5		56.3	349.4		*
Mean SiteB		N=16		35.9	110.8	14			34.1	233.9	14	42.6	201.6	11	
Mean both Sites		N=31							33.0	233.0	9	41.9	200.0	8	

* VGP not included in the means. Jsrm/Jnrm: relative intensity of the srm. Between brackets, number of specimens processed for each sample.

the contact between them being an erosion surface carved in the plutonic rocks. The Choiyoi Fm. consists of a thick sequence of andesites, dacites, rhyolites, volcanic breccias and tuffs, with thin intercalations of epiclastic rocks near the base. Most authors have assigned this unit to the Permo-Triassic "sensu lato" by stratigraphic relations and geological correlations, but no radiometric ages have been reported. In this region, the Choiyoi Fm. has an extremely variable thickness, with the thickest exposed sequence - 867 meters (H.Leanza, Servicio Geologico Nacional, unpublished report, 1987) - at the Cerro Chachil (site A, Fig.7). The Choiyoi Fm. is overlain by the volcaniclastic sequence of the Lapa Fm. (Groeber, 1956) which has been assigned to the Upper Triassic - Lower Sinemurian (Leanza, 1987); a slight uncomformity being recognized between them. This sequence is, in turn, uncomformably overlain by the thick Jurassic and Cretaceous marine and continental sedimentary sequences that make up the Neuquen Basin fill.

Although this area has a complex tectonic history, the Choiyoi rocks are always exposed as tilted fault blocks with little internal deformation.

Paleomagnetic study. The sampling of the Choiyoi rocks was carried out at two sites (A and B, Fig.7) separated by a distance of about twenty km. Ninety hand-samples, the orientation of which was measured using magnetic and sun compasses, were collected from sixty two different flows (or beds) at site A. Here, the sampling was made in an almost continuous and complete sequence, tilted to the east and exposed near the headwaters of the arroyo Nireco. A complementary sampling was carried out on a more limited section of the same sequence at site B, on the southeastern flank of the Sierra de Catan Lil. Twenty two hand-samples were collected from sixteen south-dipping flows at this site.

Twenty three pilot specimens were selected from those collected at site A and then submitted to alternating field demagnetization. The first cleaning

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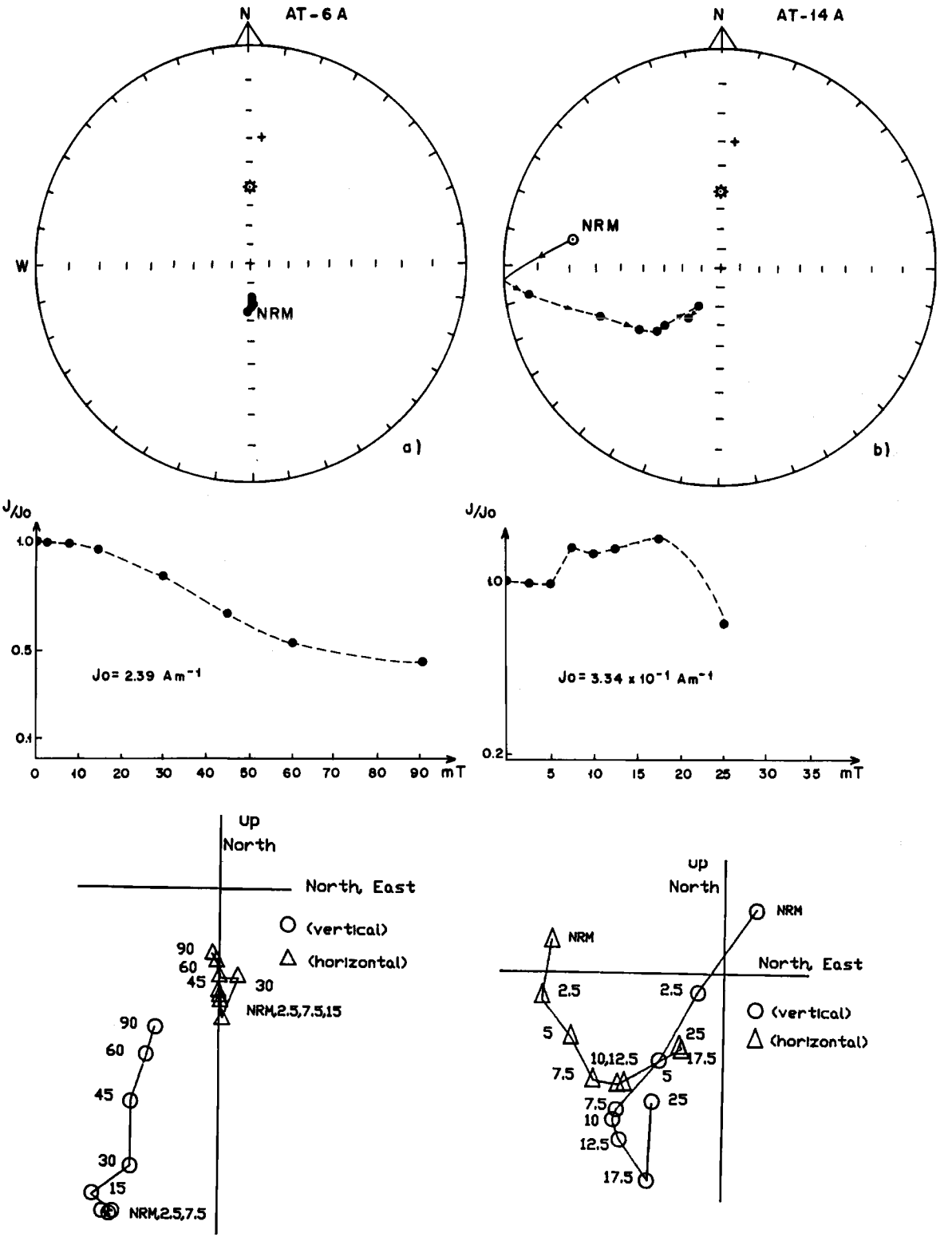


Fig.5. Typical magnetic behaviours of sspecimens from the Portezuelo del Cenizo Formation submitted to alternating field demagnetization. Magnetic directions are not corrected for tilt of the flows. Same symbols as in Fig.3.

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TABLE 2. Paleomagnetic data for the Portezuelo del Cenizo Formation

	sample No	Flow No	Jsrm/ Jnrm	Selec. stage (mT)	SRM Inc. (o)	SRM Dec. (o)	A95 (o)	Attitude strike (o)	Attitude dip	Pre-folding SRM Inc. (o)	Pre-folding SRM Dec. (o)	Pre-folding SRM A95 (o)	Lat.S (o)	VGP Long.E (o)	VGP A95 (o)	
Site C	1-2(4)	8	0.55	45	56.7	124.5		186	51 W	64.3	238.9		43.4	235.7		
	3-4(4)	7	0.45	45	66.7	142.7				52.2	248.1		33.6	221.0		
	5(2)	6	0.52	45	74.2	161.2				44.0	255.9		24.7	216.5		
	6-7(4)	5	0.51	60	66.6	171.1				41.1	245.3		32.4	209.7		
	8(2)	4	0.52	60	80.4	311.1				31.0	282.4		-1.2	221.2		
	9-10(4)	3	0.48	45	69.0	203.5				30.2	252.7		23.0	205.8		
	11/13(6)	2	0.82	17	53.1	202.8				21.6	237.8		33.1	192.7		
	14-15(4)	1	0.82	25	62.2	210.8				23.9	248.4		24.8	200.0		
Mean data site C		N=8			71.3	177.6	13			39.2	251.8	13	27.4	212.5	12	
Site B	123/125(5)	10	0.70	20	37.0	215.0		232	30 N	39.6	239.4		37.0	207.0		
	122(2)	9	0.81	20	28.0	206.0				36.9	224.8		49.0	197.0		
	120-121(3)	8	0.72	20	32.0	204.0				41.1	225.9		49.0	202.0		
	119(2)	7	0.60	20	41.0	231.0				35.1	254.8		23.0	211.0		
	118(2)	6	0.72	25	-43.0	334.0				-71.5	350.6		-64.0	125.0		*
	116-117(4)	5	0.70	20	48.0	217.0				46.9	250.9		30.0	218.0		
	112/114(6)	4	0.67	20	48.0	239.0				37.1	265.7		-14.0	217.0		
	110-111(4)	3	0.85	20	48.0	237.0				37.9	264.3		15.0	217.0		
	108-109(3)	2	0.16	40	35.0	224.0				33.6	245.0		31.0	205.0		
	107(2)	1	0.81	20	45.0	207.0				49.6	240.4		39.0	218.0		
Mean data site B		N=9			40.9	219.2	8			40.6	245.9	8	32.0	211.0	9	
Mean data both sites		N=17								40.0	248.7	7	30.0	212.0	7	

Data from site B (Middle member) were obtained by Vilas and Valencio (1982). * VGP not included in the means. Between brackets, number of specimens processed for each flow.

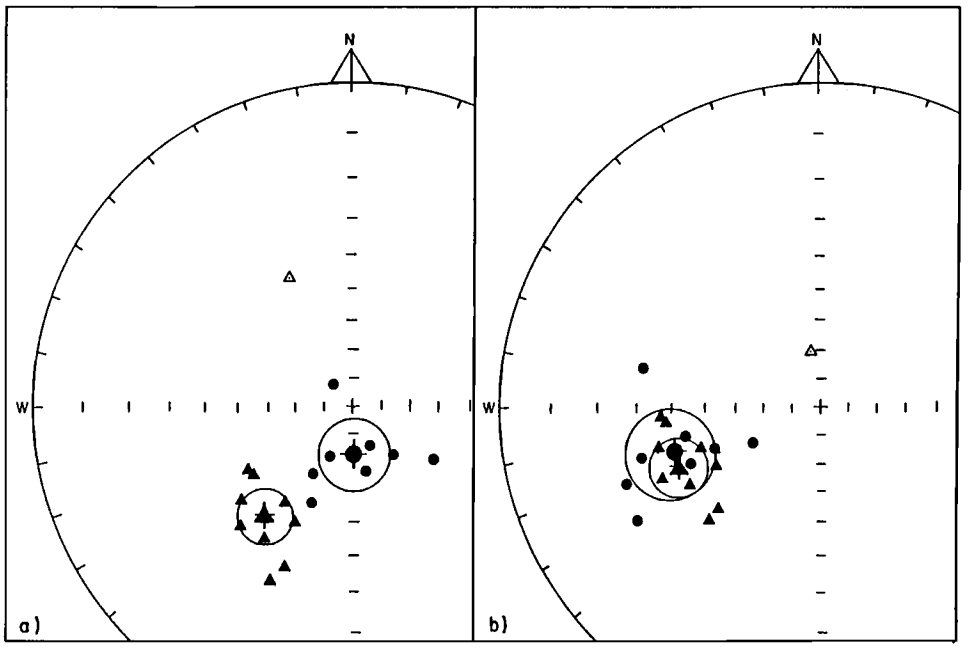


Fig.6. a) Directions of the srm without structural correction for flows (or pyroclastic beds) from the Portezuelo del Cenizo Formation. Circles: upper member; triangles: middle member; crossed circle and triangle: mean directions. Open triangle in upper hemisphere are not used in calculation of the mean. b) Directions of the srm after the application of the structural correction.

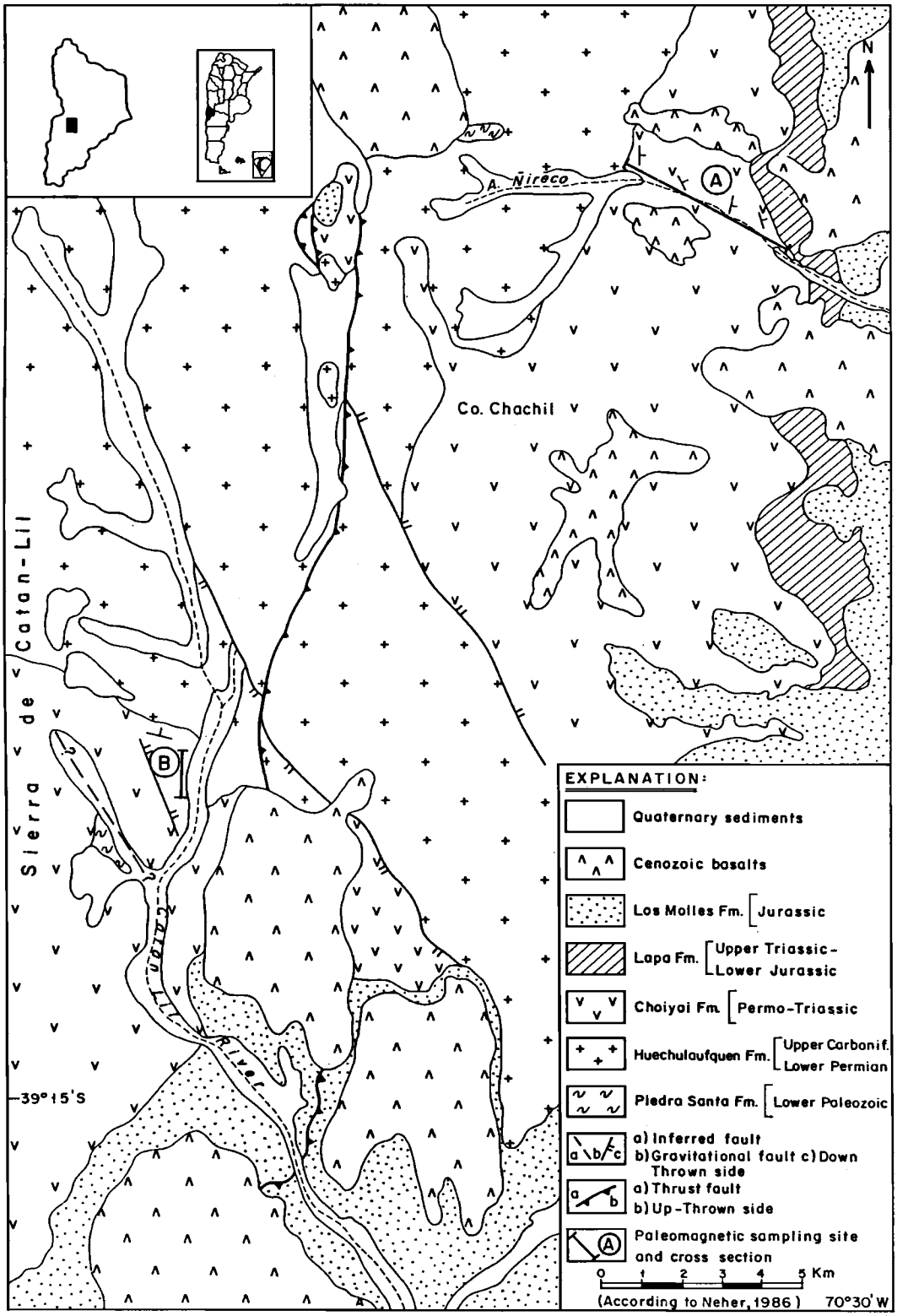


Fig.7. Geologic map of the Cerro Chachil region (sampling area 3) and the location of the sampling sites A and B.

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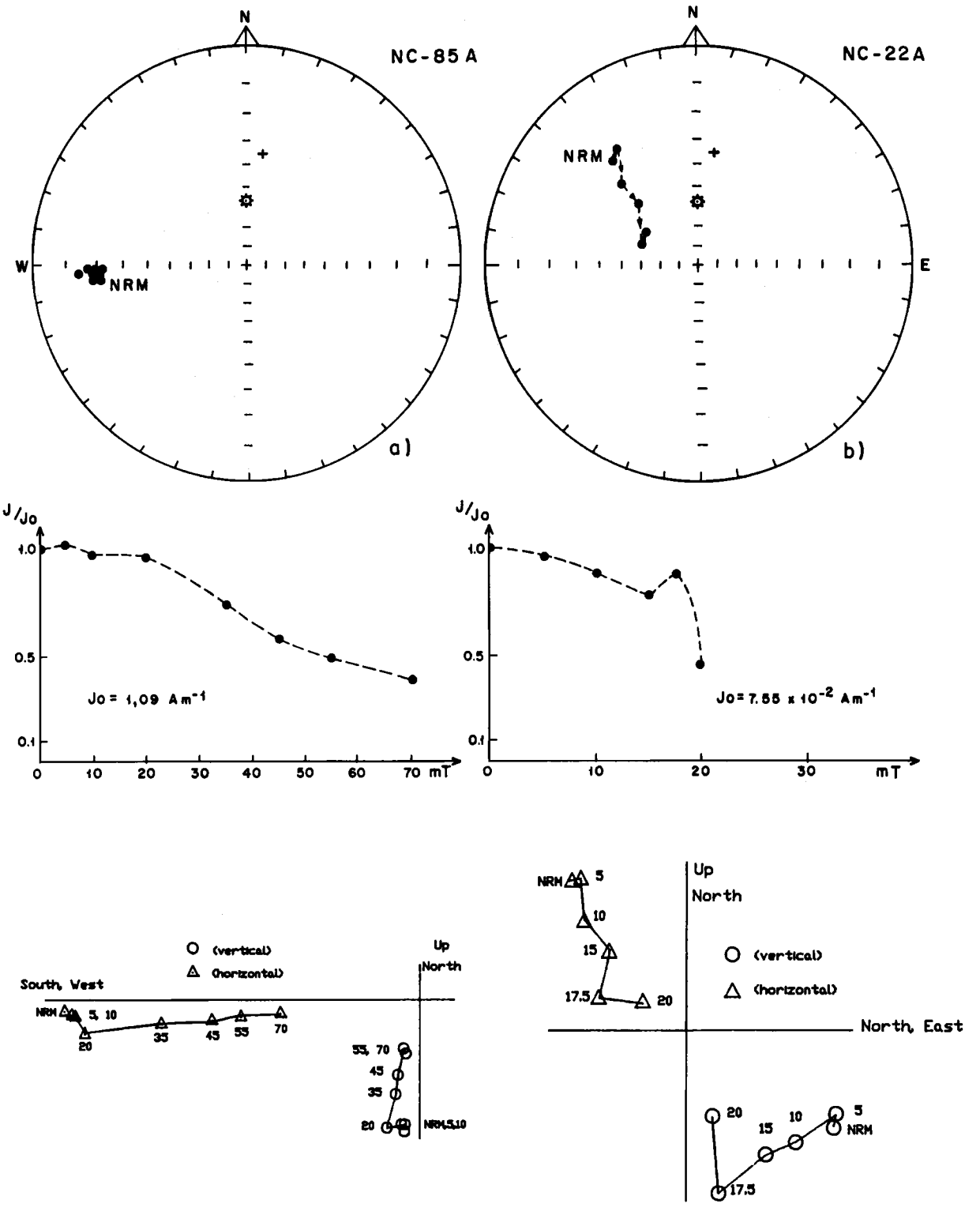


Fig.8. Typical magnetic behaviour of samples from the Choiyoi Formation collected at site A submitted to alternating field demagnetization. a) Univectorial sample (UV); b) Multivectorial sample (MV). Symbols as in Fig.3.

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step was always 5 mT, the next steps varied according to the magnetic behaviour of the specimens. Eight to ten stages were generally applied. In general, the highest field applied was 60 mT. The cleaning steps for the remaining specimens (in most cases two from each hand-sample) were determined on the basis of the magnetic behaviour of the pilot specimens. In this way, a srm could be isolated for most of the processed samples. Twenty six samples collected at site A had to be discarded because they did not carry a srm or did not show within sample (or within flow) consistency of direction of the srm. According to the magnetic behaviour shown, the remaining samples can be separated into two groups. Most samples carried only a single and very stable magnetic component with high coercivities (univectorial samples, UV; e.g. specimen NC-85A, Fig.8a). The other behaviour, less common, was only present in the lower nine tuff beds, and in four consecutive beds in a central section of the sequence. These were characterized by two superimposed magnetic components (multivectorial samples, MV). One, relatively coincident with the direction of the axial dipole, was easily cleaned by alternating field demagnetization. The other was a stable component with moderate coercivities and a positive inclination not coincident with the direction of the axial dipole (e.g. specimen NC-22A, Fig.8b).

Other ten specimens from site A were submitted to thermal demagnetization. In all cases the direction of the srm isolated by this treatment was coincident with that isolated by AF demagnetization. Curie points found for them were distributed between 575°C and 675°C. This, together with the observation of some petrographic thin-sections, sugggest that minerals belonging to the ilmenite - hematite series were the dominant ferromagnetic phases.

The twenty two hand-samples collected at site B - in most cases two specimens from each sample - were submitted to AF demagnetization. Eleven of them showed only a single magnetic component coincident with the direction of the axial dipole (e.g. specimen CL-3AB, Fig.9a). On the other hand, four samples were carriers of a single magnetic component with

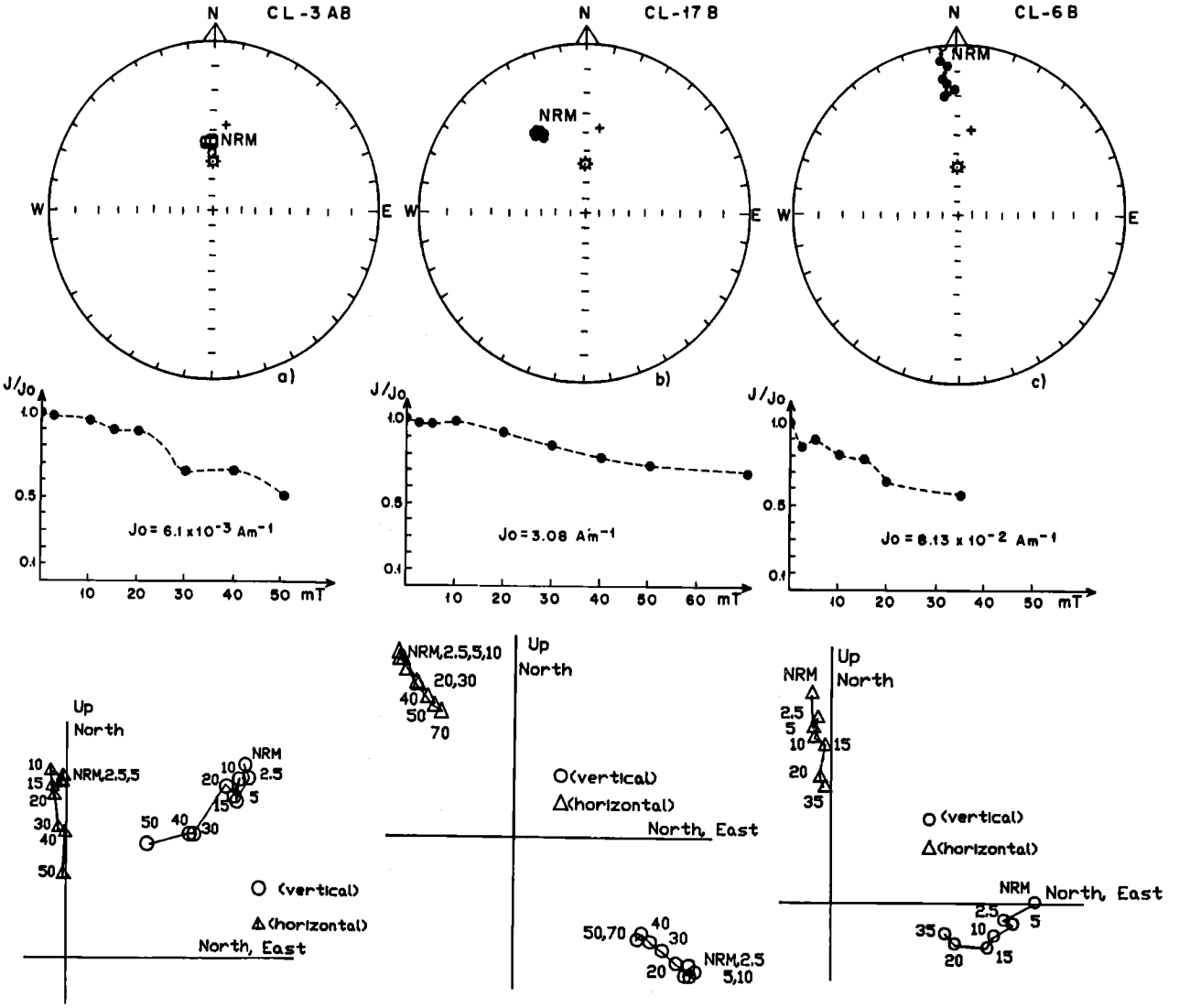


Fig.9. Typical magnetic behaviour of samples from the Choiyoi Formation collected at site B submitted to alternating field demagnetization. Symbols as in Fig.3

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TABLE 3. Paleomagnetic data for the Choiyoi Formation

	sample No	flow No	Jsrm/ Jnrm (mT)	Selec. stage	SRM Inc. (o)	SRM Dec. (o)	A95 (o)	Attitude strike (o)	Attitude dip (o)	Pre-folding Inc. (o)	Pre-folding Dec. (o)	SRM A95 (o)	Lat.S (o)	VGP Long.E (o)	VGP A95 (o)	
Co.	18(2)	5	0.75	25	66.8	263.9		4	42 E	70.4	105.8		39.5	335.8		#
Chachil	19(2)	6	0.80	15	43.4	307.3				66.5	5.2		-1.8	292.9		#
(site A)	22(2)	9	0.47	20	58.9	301.1				70.7	48.5		12.8	315.6		#
	25(1)	10	0.40	35	57.3	255.2				75.4	137.6		55.5	322.9		#
	28(2)	12	0.52	22	47.1	251.9				75.1	179.2		67.1	290.4		#
	30(2)	13	0.40	25	42.7	284.6				80.9	332.4		23.0	280.6		#
	31(2)	14	0.23	22	56.9	275.0				81.1	90.9		37.3	311.5		#
	32(2)	15	0.60	30	49.2	273.8				88.8	100.2		39.5	292.5		#
	33(2)	16	0.38	20	56.2	276.2				81.7	85.4		36.0	309.7		#
	38-39(4)	20	0.68	40	8.3	270.7		323	52 E	44.9	291.7		1.4	233.0		
	40-41(3)	21	0.78	35	16.7	279.6				44.1	308.7		-9.3	244.1		
	43(2)	23	0.71	65	6.8	254.3				53.3	270.0		20.6	227.0		
	44(2)	24	0.56	20	-53.0	269.7				-16.2	267.3		-3.1	191.4		*
	45(2)	25	0.70	70	2.5	231.1				54.5	229.7		50.6	209.6		
	47(2)	26	0.88	50	16.4	242.2				66.9	256.0		37.7	237.4		
	48(2)	27	0.75	70	1.8	275.7				36.7	290.7		-2.1	228.2		
	50(2)	29	0.80	30	24.7	276.3				51.1	315.9		-8.1	252.8		
	51(1)	30	1.08	25	55.1	251.5				68.8	22.8		3.4	303.2		#
	52(2)	31	0.43	12	51.9	243.8				74.1	28.0		12.1	303.2		#
	54(2)	32	0.98	20	44.8	254.5				72.5	352.8		7.1	285.6		#
	55-56(3)	33	0.70	40	56.8	212.1				66.7	82.6		24.4	334.7		#
	57(2)	34	0.74	35	18.0	246.8				66.6	267.9		29.7	240.6		
	58(2)	35	0.40	+	15.8	217.8				64.1	197.8		75.3	232.5		*
	59(2)	36	0.05	+	10.2	232.1				62.2	231.1		52.0	223.1		
	61(2)	38	0.96	20	19.6	100.5				-17.2	99.6		1.8	32.4		
	62(2)	39	1.12	60	40.9	217.4				77.6	124.6		49.0	319.8		*
	63(2)	40	1.00	60	11.3	268.7				48.5	292.6		2.9	235.9		
	64(2)	41	0.73	45	26.7	214.6				70.8	173.8		73.4	302.0		*
	65(2)	42	0.31	30	31.9	275.3		17	34 E	64.5	263.3		31.3	236.2		
	66(1)	43	0.94	75	25.0	271.1				56.9	259.9		29.5	225.7		
	67-68(3)	44	0.56	60	15.5	257.8				43.8	246.0		4.0	33.6		
	8(2)	47	0.95	60	12.8	272.9				45.5	267.2		18.7	219.5		
	9-10(4)	48	1.14	20	38.1	243.9		343	29 E	66.4	234.9		50.2	232.2		
	90/92(6)	49	0.80	45	55.5	299.3				73.5	357.7		8.5	288.3		*
	89(2)	50	0.84	60	16.7	266.0				44.7	270.7		15.7	220.9		
	87-88(4)	51	0.58	40	7.9	251.1				36.9	250.5		27.7	204.4		
	83/86(7)	52	0.70	50	25.5	252.6				54.4	252.4		33.6	219.7		
	82(2)	53	0.88	50	39.4	255.2				68.4	257.6		36.7	240.3		
	81(2)	54	0.68	40	26.6	267.5				54.2	275.5		17.4	230.5		
	80(2)	55	0.83	55	40.3	48.8				28.9	54.3		-15.6	343.8		*
	79(2)	56	0.83	50	2.1	151.0				-3.9	151.5		41.3	70.1		*
	76-77(4)	58	0.80	50	28.7	255.7				57.6	257.5		31.5	225.4		
	75(2)	59	0.71	45	36.0	251.2				65.0	249.6		40.2	232.6		
	73-74(4)	60	0.56	45	56.3	60.4				27.7	65.1		-9.0	352.1		*
	71-72(4)	61	0.82	50	-0.6	281.9				24.5	285.1		-3.3	218.9		
	69-70(3)	62	0.65	50	-17.0	286.8				7.3	285.4		-9.5	212.2		
mean UV samples		N=26			16.6	261.7	8			52.1	269.9	7	22.2	226.9	8	
mean MVsamples		N=13			55.3	266.3	8			80.8	55.1	7	28.3	305.6	14	
Sa.	6(2)	5	0.57	35	20.8	355.2		103	53 S	67.6	324.4		5.0	267.5		
CatanLil	7(2)	6	0,54	30	21,2	337.6				55.6	300.2		3.3	244.9		
(site B)	10(2)	8	0.29	25	26.9	359.9				75.0	321.7		15.7	271.6		
	15(2)	11	0.39	20	31.8	323.5				49.3	275.2		14.9	226.3		
	16-17(4)	12	0.72	70	34.6	337.0				60.9	276.7		21.2	237.9		
	21(2)	15	0.50	70	28.9	3.5				78.7	325.8		20.4	276.4		
	22(2)	16	0.98	40	34.9	335.3				59.7	275.2		20.9	235.2		
mean site B		N= 7			29.1	344.9	11			65.3	293.2	11	15.2	251.4	16	
mean both sites		N= 33								53.5	273.8	8	21.0	232.28	8	

Univectorial samples, not included in the mean for both sites. * VGP not included in the means. + srm isolated by thermal cleaning. Between brackets, specimens processed for each flow.

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a positive inclination, not coincident with the direction of the dipole. This component showed high coercivities (e.g. specimen CL-17B, Fig.9b). The remaining nine samples showed a superposition of both components (e.g. specimen CL-6B, Fig.9c). The coincidence of the "in situ" direction of the first mentioned component with that of the axial dipole indicates that it corresponds to a secondary magnetization, probably due to the intrusion of some Late Cenozoic basaltic dykes located near site B. The other component could be isolated for only eight samples from seven different flows (Table 3).

The srm directions for all samples from each flow of the Choiyoi Fm. were averaged in order to obtain a single srm direction for each flow. Sequences sampled at sites A and B have clearly different attitudes, allowing the application of a tilt test. Fig. 10a shows the srm directions and the mean directions for sites A and B before structural corrections are applied. Directions from site A have been plotted with different symbols according to the magnetic behaviour they showed (UV samples: circles; MV samples: squares). MV directions seem to be grouped in a different population than the UV directions, although the dispersion of the latter is important. If we consider only the UV population from site A, the mean in situ srm directions for sites A and B are clearly different (Fig.10a). Instead, after the structural corrections, both mean directions overlap (Fig.10b). This indicates that the srm isolated from both sites is pretilting. On the other hand, MV directions from site A (marked with # in Table 3) do not pass the tilt test, suggesting that they may be secondary magnetizations (Rapalini, 1988). For this reason they were not taken into account in the calculation of the paleomagnetic pole.

A VGP for each flow or bed that carried a pretilting srm was computed (Table 3). The mean of these is the paleomagnetic pole for the Choiyoi Fm. (P14): 232°E, 21°S, N=33, k=11.7, A95=8°. Similarly as for the Majaditas and Portezuelo del Cenizo Formations, all VGPs located beyond 40° from the mean or its antipode were considered recordings of transitional fields or excursions (marked with * in Table 3) and excluded from the mean.

Discussion and Interpretations

Figure 11 shows the Late Paleozoic apparent polar wandering path (apwp) for the stable South American craton and some of the paleomagnetic poles that contribute to it (Table 4). The location of the paleomagnetic poles for the Majaditas Fm. (C8), the Portezuelo del Cenizo Fm. (P13), the Choiyoi Fm. (P14) and the Yalguaraz Fm. (C7, Vilas and Valencio, 1982) are also shown. C7, C8, P13 and P14 do not agree with the apwp for South America. These "anomalous" pole positions are interpreted in terms of two different geodynamic hypotheses.

Hypothesis A

This hypothesis postulates that sampling areas 1, 2 and 3 (see Figs. 2 and 7) are situated in crustal blocks that suffered large "in situ" clockwise rotations around vertical axes during the Mesozoic or Cenozoic.

Figure 11 shows the positions of these "anomalous paleomagnetic poles" and how they are rotated counterclockwise around vertical axes located at their respective sampling areas, in order to agree with the Late Paleozoic apwp of South America. C'8 indicates the position of C8 after a counterclockwise rotation of 85°, C'7 and P'13 the positions of C7 and P13 after a rotation of 70° and P'14 the position of P14 after a counterclockwise rotation of 80°. The positions of C'7 and C'8 are relatively near the Upper Carboniferous mean paleomagnetic pole (Cu, Vilas 1981) although they are not situated on the South American path. On the other hand, P'13 and P'14 seem to agree better with the Permo-Triassic section of the apwp for the stable South American craton. Accurate values of rotation (R) and paleolatitude anomaly (P) computed following Demarest (1982) are shown in Table 5. As the ages of C7, C8, P13 and P14 are not too well constrained, these values must be considered cautiously. Nevertheless, C7 and C8 show

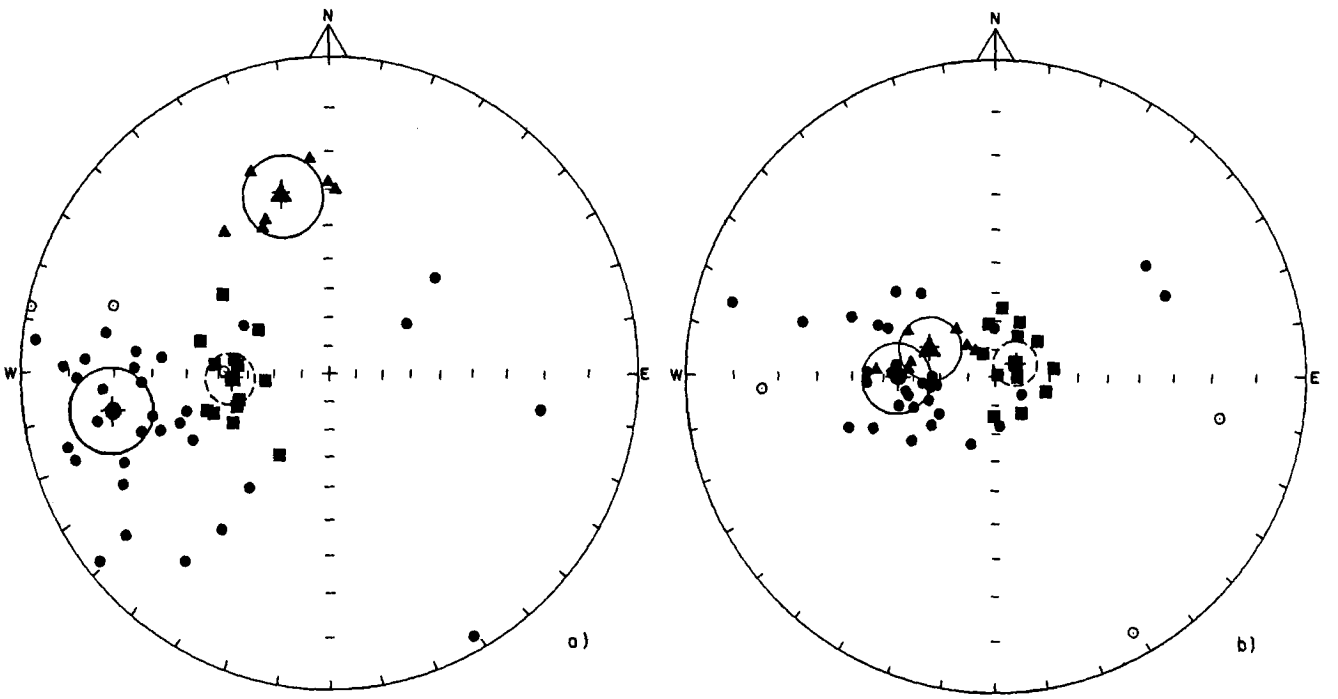


Fig.10. a) Directions of the srm without structural correction for the flows (or pyroclastic beds) from the Choiyoi Formation. Circles: flows from site A with UV behaviour; squares: flows from site A with MV behaviour; triangles: flows from site B; crossed circle, square and triangle: mean directions. b) Directions of the srm after the application of the structural corrections.

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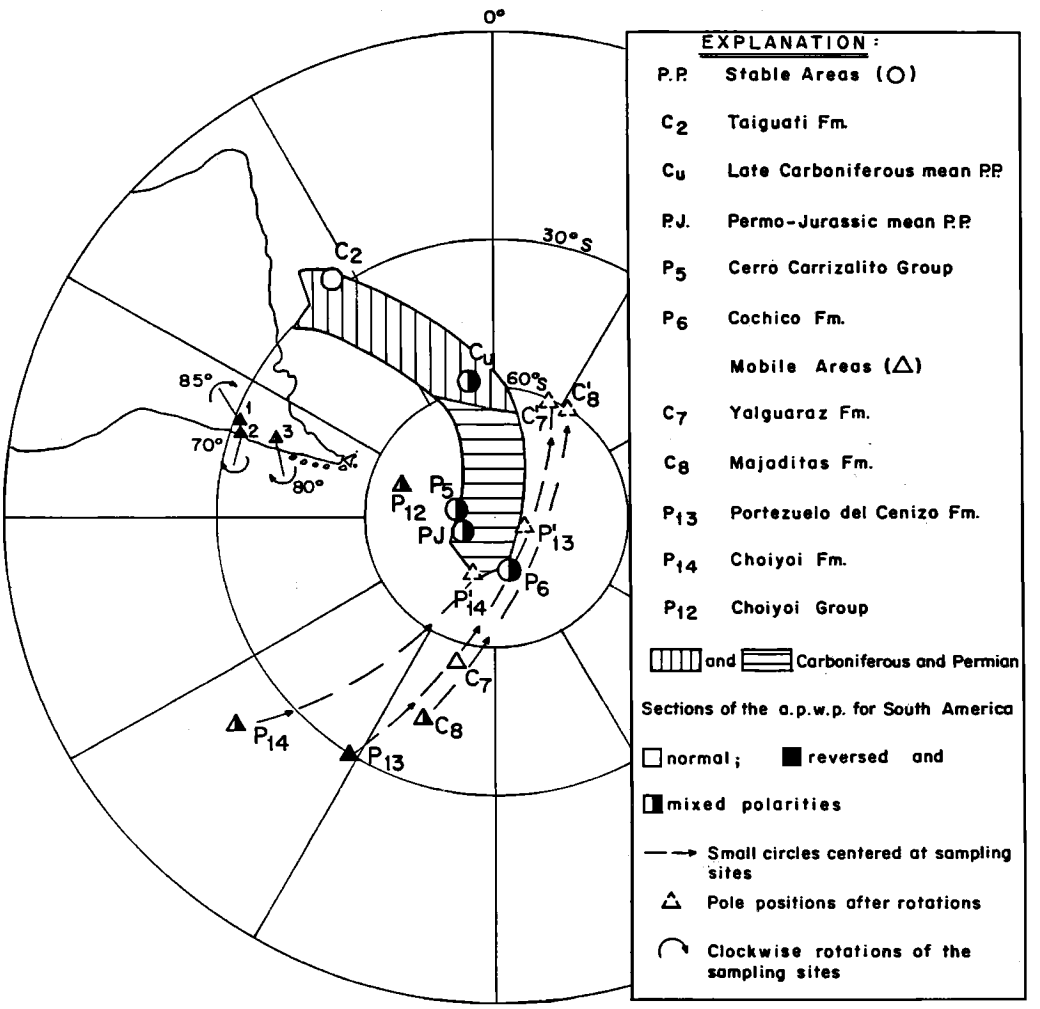


Fig.11. Hypothesis A: "Crustal block rotations". C7, C8, P13 and P14 are not coincident with the late Paleozoic section of the apwp for stable areas of South America. After different counterclockwise rotations, due to clockwise rotations of the sampling areas, they are restored to the best coincident positions with those for correlative rocks from stable areas of South America. Data in Table 4.

significant paleolatitudes anomalies that cannot be explained by the hypothesis of in situ crustal block rotations unless the poor agreement of these poles with the Upper Carboniferous mean paleomagnetic pole is attributed to a non accurate definition of this section of the South American apwp (Vilas and Valencio, 1982; Ernesto et al, 1988). The paleolatitude anomalies shown by P13 and P14 are considered not significant because they dissapear if the paleomagnetic pole of the Permian Cochicó Formation (276±5 Ma, Linares et al, 1979), is taken as the reference pole instead of the Permo-Jurassic mean paleomagnetic pole (Table 5).

On the basis of geological evidence it is assumed that sites 2 A,B and C are situated in the same rotated crustal block, being a value of rotation of about 70 degrees compatible with the rotations computed from C7 and P13. Then, this hypothesis suggests large in situ clockwise rotations of about 85° for sampling area 1, of about 70° for sampling area 2 and near 80° for sampling area 3.

The paleomagnetic studies reported here were carried out, in each case, on two sequences of the same formation with different attitudes. This rules out the existence of "apparent rotations" caused by two or more superimposed non controlled tectonic movements (MacDonald, 1980). Therefore, the postulated rotations must be considered true tectonic rotations of crustal blocks.

Neher (1986) has recently accomplished a detailed structural study of the Cerro Chachil region. Some of his conclusions support the rotation hypothesis for sampling area 3. According to Neher (1986), there is evidence that this region was affected by a significant shear stress which caused large dextral displacements along east-west trending strike-slip faults. These movements might have been related to the development of a major E-W positive structural feature in the Neuquen Basin, called the "Neuquen Dorsal". This feature is considered to have been developed by dextral transpressive movements mainly in the Late Jurassic and/or in the Middle Cretaceous (Orchuela et al, 1981; Bettini, 1984). Thus, the proposed rotation for the Cerro Chachil probably happened at those times. In the presence of a regional shear stress, this crustal block may have behaved rigidly, rotating with little internal deformation, similar to the model proposed by Beck (1980). The boundaries of the crustal block under discussion cannot yet be defined. Nevertheless, a length of at least 20 km - the distance between both sampling sites - can be inferred for it.

Structural studies for the remaining sampling areas are not so detailed.

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TABLE 4. Data for the paleomagnetic poles shown in Figs. 11 and 12.

	Age	P.P. Lat.S (o)	P.P. Long.E (o)	A95 (o)	Symbol	References
Stable zones						
Permo-Jurassic mean PP for South America	Late Permian-Mid Jurassic	58	350	6	PJ	Vilas (1981)
Cochicó Formation	276±5 Ma	77	163	5	P6	Vilas & Valencio (1982)
Cerro Carrizalito Group	263±5 Ma	81	282	5	P5	Valencio & Mitchell (1972)
Late Carboniferous mean PP for South America	Late Carbonif.	58	350	6	Cu	Vilas (1981)
Taiguati Formation	Moscovian	28	326	8	C2	Creer (1965)
Probably mobile zones						
Ignimbrites from the Choiyoi Group	228±10 Ma	67	290	10	P12	Vilas & Valencio (1982)
Choiyoi Fm.	Permo-Triassic	21	232	8	P14	This paper
Portezuelo del Cenizo Fm.	275±10 Ma	30	212	7	P13	This paper
Yalguaraz Fm.	Late Carbonif.	56	195	7	C7	Vilas & Valencio (1982)
Majaditas Fm.	Westphalian (?)	42	200	8	C8	This paper

This prevents us from suggesting any tectonic mechanism for the proposed rotations. Nevertheless, the Precordillera as well as the Cordillera Frontal have complex tectonic histories (Caminos, 1979; Furque and Cuerda, 1979) in which "in situ" crustal block rotations around vertical axes could have taken place.

Hypothesis B

According to this interpretation, C7, C8, P13 and P14 define a Late Paleozoic apwp different from that established for stable South America (Fig. 12a). This means that the sampling areas 1, 2 and 3 would form part of a displaced terrane.

P12 is the PP for a sequence of ignimbrites of the Choiyoi Group exposed in Quebrada de La Alumbrera, Cordillera Frontal (31.6°S, 69.5°W, San Juan Province, Argentina; see area 4 in Fig. 1). A K/Ar radiometric date of 228±10 m.a. has been reported for these rocks (Vilas and Valencio, 1982). The exposures of these ignimbrites are not too distant from those of the Portezuelo del Cenizo Fm., which also forms part of the Choiyoi Group (Cortés, 1985), and both are situated in similar geologic settings. Thus, it is reasonable to conclude that these ignimbrites (area 4) must be situated in the displaced terrane too. As the position of P12 is consistent with those reported for correlative rocks from stable areas of South America we infer that both apwp became coincident before 228±10 m.a (Fig. 12a). This means that the proposed displaced terrane must have been already accreted to the western edge of South America by the Early Triassic.

Available paleomagnetic data for Andean Late Paleozoic rocks are still quite scarce. Thus, we cannot suggest the location of the limits of the displaced terrane. Nevertheless, its eastward extension is limited by the San Rafael morphotectonic province and the Paganzo Basin, the Carboniferous and Permian PPs of which are consistent with those from stable areas of South America (Vilas, 1981).

Figure 12b shows a schematic paleogeographic reconstruction for the proposed displaced terrane and South America during the Late Paleozoic. The limits drawn for the terrane are only diagrammatic and have no actual significance. This hypothetic reconstruction was done to provide the best overlap of their respective apwp. This reconstruction also suggests that the displaced terrane might have been situated at subtropical latitudes during the Late Carboniferous.

From figures 11 and 12 it clearly follows that C7 and C8 harmonize better with the apwp for South America in hypothesis B. Nevertheless, the first interpretation seems to be better supported by known geological evidence. On one hand, the transcurrent movements reported for the sampling area of the Choiyoi Fm. may have caused an "in situ" clockwise rotation of that area. On the other, geological evidence known up to date for this section of the Southern Andes does not seem to support the existence of a Late Paleozoic allochthonous terrane like the one postulated here (Ramos et al, 1986).

TABLE 5. Values of rotation (R), paleolatitude anomalies (P) and the respective errors (ER and EP) computed for the Majaditas, Yalguaraz, Portezuelo del Cenizo and Choiyoi Formations.

Paleomagnetic pole	Reference pole	R	ER	P	EP
C7	Cu	75.3	± 8.9	20.0	± 7.4
C8	Cu	88.7	± 9.4	24.6	± 8.0
P13	PJ	61.3	± 7.3	15.4	± 6.4
	P6	57.0	± 7.5	0.9	± 6.9
P14	PJ	82.2	± 9.0	11.2	± 7.2
	P6	78.8	± 9.0	-3.3	± 7.5

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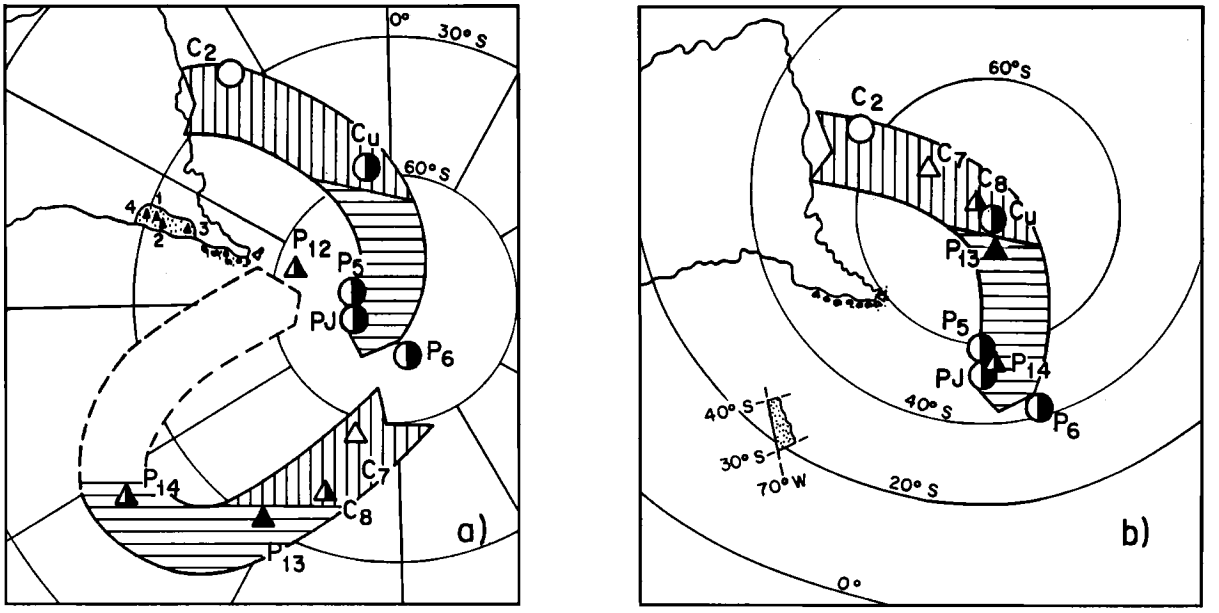


Fig.12. Hypothesis B: "A displaced terrane". a) The sampling areas for the Majaditas Fm. (1), Yalguaraz Fm. (2), Portezuelo del Cenizo Fm. (2), Choiyoi Fm. (3) and Ignimbrites from the Choiyoi Group (4) might be situated in a displaced terrane. The paleomagnetic poles for these rocks (C8, C7, P13, P14 and P12 respectively) seem to define an apwp different from that for South America. b) Schematic paleogeographic reconstruction for the displaced terrane and South America during the late Carboniferous-early Permian, obtained by the best overlap of the late Paleozoic sections of their apwp. Paleolatitudes are valid for the late Carboniferous. Data in Table 4.

The lack of further paleomagnetic data for Late Paleozoic rocks from the Southern Andes as well as some vagueness in the age assigned to some of the studied formations do not allow us to favor one hypothesis over the other. Furthermore, they may be considered as two extreme hypotheses. A combination of both, like for example: an "in situ" clockwise rotation of the Cerro Chachil area together with the existence of a smaller displaced terrane integrated by the sampling areas 1, 2 and 4 is also possible.

Conclusions

Paleomagnetic data for rocks from the Majaditas Fm., the Portezuelo del Cenizo Fm. and the Choiyoi Fm., all assigned to the Late Paleozoic and exposed in different areas of the Argentine Andean Belt, indicate that these rocks carry a primary remanent magnetization.

Paleomagnetic pole positions computed for these formations are not consistent with the Late Paleozoic apparent polar wandering path for South America. This was interpreted in terms of two different hypotheses. The first postulates that the sampling sites for these rocks are located in crustal blocks that suffered large tectonic rotations (70° to 85°, clockwise) around vertical axes, probably during the Mesozoic or Cenozoic. The other suggests that all these formations form part of a Late Paleozoic displaced terrane, the limits of which are not defined yet. This terrane should have been accreted to the western edge of South America prior to the Early Triassic.

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